A

Compound class distribution

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	other
MatterGen Base	48.8	14.8	0.9	1.7	2.1	4.2	2.1	3.8	2.5	4.6	14.5
MatterGen FT	55.0	0.0	1.6	0.1	17.4	0.0	2.5	5.4	12.8	0.0	5.1
Con-CDVAE Base	14.4	21.0	1.9	14.4	1.6	14.0	0.3	4.3	0.7	13.5	13.8
Con-CDVAE FT	18.6	0.0	4.8	1.8	14.2	0.0	27.7	9.5	6.3	0.0	16.9
CrystalFormer Base	28.9	11.7	24.5	8.8	2.4	4.2	0.7	3.3	2.0	0.4	13.1
CrystalFormer FT	45.4	1.9	14.6	1.2	11.2	2.2	2.8	5.4	8.7	0.2	6.4
CrystaLLM Base	29.6	45.9	0.0	0.0	0.0	12.2	8.2	0.0	4.1	0.0	0.0
CrystaLLM FT	74.2	0.0	2.3	0.9	0.0	0.5	3.2	0.5	18.4	0.0	0.0
Crys-JEPA Base	30.3	16.3	1.7	13.2	4.7	11.3	2.2	4.0	5.0	3.1	8.3
Crys-JEPA Refined	76.1	0.0	20.2	0.3	0.0	0.1	0.4	0.9	0.1	0.0	1.9

Share within cohort (%)

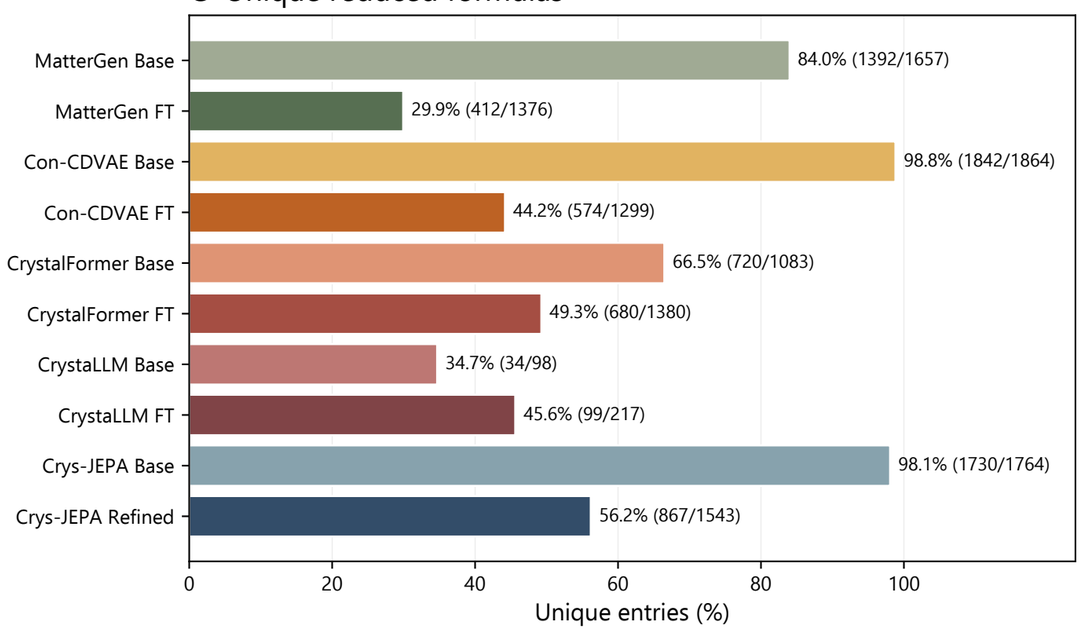
B

Changes associated with adaptation

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	other
MatterGen	+6.2	−14.8	+0.7	−1.5	+15.3	−4.2	+0.4	+1.6	+10.3	−4.6	−9.5
Con-CDVAE	+4.2	−21.0	+2.9	−12.7	+12.6	−14.0	+27.4	+5.3	+5.6	−13.5	+3.1
CrystalFormer	+16.5	−9.8	−9.8	−7.5	+8.8	−1.9	+2.0	+2.0	+6.7	−0.2	−6.7
CrystaLLM	+44.6	−45.9	+2.3	+0.9	+0.0	−11.8	−4.9	+0.5	+14.4	+0.0	+0.0
Crys-JEPA	+45.8	−16.3	+18.5	−12.8	−4.7	−11.2	−1.8	−3.1	−4.9	−3.1	−6.4

Fine-tuned/Refined minus Base (pp)

C Unique reduced formulas



D Unique formula–space group pairs

